

Lecture 14: The number of Sidon sets

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1.1 The number of Sidon sets

A set $A \subseteq [n]$ is a *Sidon set* if there is no *Sidon 4-tuple*, i.e., (a, b, c, d) in A with $a + b = c + d$ and $\{a, b\} \cap \{c, d\} = \emptyset$. For example, $(4, 4, 5, 3)$ is a Sidon 4-tuple since $4 + 4 = 5 + 3$. A major problem concerning Sidon sets is finding the largest cardinality of a Sidon set contained in $[n]$. Let $\xi(n)$ be the maximum size of a Sidon set A with $A \subseteq [n]$. We have a trivial bound

$$n^{1/3} \leq \xi(n) \leq 2\sqrt{n}.$$

The upper bound follows by observing that in a Sidon set A , the difference of the numbers in A are all distinct, i.e. $\binom{|A|}{2} \leq n$. The lower bound is obtained by using the greedy algorithm.

By the works of Erdős and Turán [6], Singer [7], Erdős [4], and Chowla [3], it is known that the maximum Sidon set is of size $(1 + o(1))\sqrt{n}$. This is also true for the Sidon sets in $\mathbb{Z}_n$, the integers modulo n .

1.1.1 A connection to C_4 -free graphs

Due to Brown [1] and Erdős Rényi, and Sós [5], it is well-known that the maximum number of edges in a C_4 -free graph is $\left(\frac{1}{2} + o(1)\right)n^{3/2}$. It is relatively easy to prove the upper bound. However, it is difficult to construct n -vertex graphs without 4-cycles with $(1/2 + o(1))n^{3/2}$ edges. For the bipartite version of the problem, the same order of magnitude is obtained by Reiman [9]. More precisely, he showed that

$$\text{ex}(n, n, C_4) = (1 + o(1))n^{3/2}.$$

Next, we give a proof of the upper bound for the bipartite version and show a construction for the lower bound of both versions.

Proof of the upper bound for bipartite version. Let G be an (A, B) -bigraph with $|A| = |B| = n$ without 4-cycles. Let $d_1, d_2, \dots, d_n$ be the degrees of vertices in A . Since G is C_4 -free, we have

$$\sum_{i=1}^n \binom{d_i}{2} \leq \binom{n}{2}.$$

Indeed, otherwise G contains a C_4 , by the pigeonhole principle. Now, noting that $\binom{x}{2}$ is a convex function, using Jensen inequality, we get

$$\frac{n^2}{2} \geq \binom{n}{2} \geq \sum_{i=1}^n \binom{d_i}{2} \geq \binom{(\sum_{i=1}^n d_i)/n}{2} = \frac{e(G)}{2n} \left(\frac{e(G)}{n} - 1 \right).$$

This implies that $e(G) \leq n^{3/2} + n/2$. ■

Construction 1.1.1 (The incidence graph of a finite projective plane, Reiman [9]). Let q be a prime power. It is known that there exists a finite projective plane with a point set $\mathcal{P}$ and a line set $\mathcal{L}$ such that the following hold:

- $|\mathcal{P}| = |\mathcal{L}| = q^2 + q + 1 =: n$,
- Each line contains exactly $q + 1$ points,
- Each point is contained in exactly $q + 1$ lines.

Let G be the $(\mathcal{P}, \mathcal{L})$ -bigraph such that there is an edge between $p \in \mathcal{P}$ and $\ell \in \mathcal{L}$ if and only if $p \in \ell$. This is an optimal C_4 -free graph. Indeed, suppose that G contains a 4-cycle $p_1 \ell_1 p_2 \ell_2$. Then we would have two different points p_1, p_2 and two different lines ℓ_1, ℓ_2 such that $p_1, p_2 \in \ell_1, \ell_2$, which is impossible. Also, the number of edges is $(q + 1)(q^2 + q + 1)$.

Construction 1.1.1 requires the existence of a projective plane with some properties. To construct such a projective plane, one can consider equivalence classes of $\mathbb{F}_q^3$, where $\mathbb{F}_q$ is the finite field with q elements. More precisely, two elements (a, b, c) and (a', b', c') in $\mathbb{F}_q^3 \setminus \{(0, 0, 0)\}$ are equivalent if $(a, b, c) = \lambda \cdot (a', b', c')$ for some $\lambda \in \mathbb{F}_q \setminus \{0\}$. We write $[a, b, c]$ for the equivalence class of (a, b, c) . Let the point set $\mathcal{P}$ be the set of equivalence classes. There are $\frac{q^3-1}{q-1} = q^2 + q + 1$ such points. For every $[a, b, c] \in \mathcal{P}$, a straight line is defined as the set of all points $(x, y, z) \in \mathbb{F}_q^3$ which satisfies the equation $ax + by + cz = 0$. We let $\mathcal{L}$ be the set of all such lines.

Construction 1.1.2 (Non-bipartite construction via projective norm-graph). Let G be a graph with vertex set $\mathcal{P}$, where $\mathcal{P}$ is the set of equivalence classes of $\mathbb{F}_q^3$ defined as above, thus $n := v(G) = |\mathcal{P}| = q^2 + q + 1$. Two different classes $[a, b, c]$ and $[x, y, z]$ are connected by an edge if and only if $ax + by + cz = 0$. One can show that G is C_4 -free graph with $e(G) = \frac{1}{2}q(q + 1)^2 = (1/2 + o(1))n^{3/2}$. We omit the details of the proof here.

Bipartite C_4 -free graphs from Sidon sets. Let $A \subset \mathbb{Z}_n$ be a Sidon set with size $(1 + o(1))\sqrt{n}$ for some odd n . We define a bipartite graph $G = (U, V)$ with vertex sets $U = V = \mathbb{Z}_n$ such that for two vertices $u \in U$ and $v \in V$, u is adjacent to v if and only if there exists a number $a \in A$ such that $u + a = v$ in $\mathbb{Z}_n$. We have $e(G) = n|A| = (1 + o(1))n^{3/2}$. We claim that G is a C_4 -free graph. Suppose that there exists a 4-cycle $x, x + a, y, y + b$ where $a, b \in A$. Since $(x, y + b)$ and $(y, x + a)$ are edges, there exist two numbers $c, d \in A$ such that $x + c = y + b$ and $y + d = x + a$. However, this also gives $c + d = a + b$, where $\{c, d\} \neq \{a, b\}$ because of $x \neq 0$ and n is odd. Therefore, we find a Sidon 4-tuple (a, b, c, d) in A , which is a contradiction.

1.1.2 The number of Sidon sets

In 1990, Cameron and Erdős [2] proposed the problem of determining the number of Sidon sets in $[n]$. By the largest cardinality of a Sidon set, it is trivial to obtain the following bound:

$$2^{(1+o(1))\sqrt{n}} \leq S(n) := \text{the number of Sidon sets} \leq \binom{n}{\leq (1+o(1))\sqrt{n}} \leq 2^{\sqrt{n} \log n}.$$

It seems natural to think that the trivial lower bound is the best possible. However, surprisingly, Saxton and Thomason [10] showed that the lower bound is not tight.

Theorem 1.1.3 (Saxton-Thomason [10]). *The number of Sidon sets in $[n]$ is at least $2^{1.16\sqrt{n}}$.*

Proof. Let n be a multiple of 4. Let $A \subseteq [n/4]$ be a Sidon set of $\mathbb{Z}_{n/4}$ with maximum size, $|A| = (\frac{1}{2} + o(1))\sqrt{n}$. For every $a \in A$, we choose u_a to be any one of $a, \frac{n}{4} + a, \frac{n}{2} + a, \frac{3n}{4} + a$, or nothing. Let $U = \{u_a : a \in A\}$. Clearly, the number of choices for U is $5^{\frac{1}{2}\sqrt{n}} \approx 2^{1.16\sqrt{n}}$. We claim that U is a Sidon set in $[n]$. Suppose that there is a Sidon 4-tuple (a', b', c', d') in U such that $a' + b' = c' + d'$. Then we can find numbers $a, b, c, d \in A$ such that $a + b \equiv c + d \pmod{n/4}$, which contradicts the fact that A is a Sidon set in $\mathbb{Z}_{n/4}$. ■

Note that the choice of $n/4$ is not crucial for the argument above, but it optimizes the lower bound obtained. We remark that Saxton and Thomason [10] obtained the upper bound $2^{(55+o(1))\sqrt{n}}$ by applying the hypergraph container method. The best result on the upper bound is given by Kohayakawa, Lee, Rödl, and Samotij [8], using the multiphase graph container method.

Theorem 1.1.4 (Kohayakawa-Lee-Rödl-Samotij [8]). *The number of Sidon sets in $[n]$ is at most $2^{c\sqrt{n}}$, where c is a constant arbitrarily close to $\log_2(33e) \approx 6.442$ for sufficiently large n .*

In this next section, we show that the number of Sidon sets is $2^{O(\sqrt{n})}$, but we do not try to optimize the constant in the exponent. We also present the proof of Theorem 1.1.4 in the last section.

1.1.3 Supersaturation

For two sets $A, U \subseteq [n]$, define a multigraph $H^U(A)$ on vertex set A such that for every $a_1, a_2 \in A$ with $a_1 < a_2$, the multiplicity of the edge a_1a_2 in $H^U(A)$ is the number of ordered pairs (u_1, u_2) in U such that $a_1 + u_1 = a_2 + u_2$. Thus the multiplicity of the edge a_1a_2 is the number of times the difference $a_2 - a_1$ appears as a difference between elements of U .

Lemma 1.1.5. *Let $A, U \subseteq [n]$. If $|A| \cdot |U| \geq 6n$, then $e(H^U(A)) > \frac{|A|^2|U|^2}{12n}$.*

Proof. Let F be a simple bipartite graph defined on parts A and $[2n]$ satisfying that for every $a \in A$ and $m \in [2n]$, a is adjacent to m if and only if there is an element $u \in U$ such that $a + u = m$. Clearly, for every vertex $a \in A$, we have $d_F(a) = |U|$.

Let $\mathcal{P}$ be the set of cherries (paths of length 2) in F with endpoints in A . Then we have

$$|\mathcal{P}| = \sum_{m \in [2n]} \binom{d_F(m)}{2} \geq 2n \binom{\frac{\sum_{m \in [2n]} d_F(m)}{2n}}{2} = 2n \binom{\frac{|A| \cdot |U|}{2n}}{2} > \frac{|A|^2|U|^2}{6n}.$$

A path $P = \{xyz\} \in \mathcal{P}$ corresponds to an edge of $H^U(A)$, except for the *trivial* paths where $x + z = y$. Note that if P is *trivial* then x and z belong to $A \cap U$. Thus, the number of trivial paths in $\mathcal{P}$ is bounded by $\binom{|A \cap U|}{2}$. Let $\mathcal{P}'$ be the set of nontrivial paths in $\mathcal{P}$. We obtain

$$e(H^U(A)) = |\mathcal{P}'| = |\mathcal{P}| - \binom{|A \cap U|}{2} > \frac{|A|^2|U|^2}{6n} - \frac{|A| \cdot |U|}{2} \geq \frac{|A|^2|U|^2}{12n},$$

where the first inequality is given by $|A \cap U| \leq \min\{|A|, |U|\} \leq \sqrt{|A| \cdot |U|}$ and the second inequality follows from the assumption $|A| \cdot |U| \geq 6n$. ■

Proposition 1.1.6. *Let $U \subseteq [n]$ be a Sidon set, and $A \subseteq [n] - U$. Then $H^U(A)$ is a simple graph.*

Proof. Suppose that $a_1 a_2$ has multiplicity at least 2. Then there exist $u_1, u_2, u_3, u_4 \in U$ such that $a_1 + u_1 = a_2 + u_2$ and $a_1 + u_3 = a_2 + u_4$ with $(u_1, u_2) \neq (u_3, u_4)$. This implies $u_1 + u_4 = u_2 + u_3$, which contradicts that U is a Sidon set. ■

Proposition 1.1.7. *Let $U \subseteq [n]$ be a Sidon set, and $I \subseteq A \subseteq [n] - U$ such that $U \cup I$ is also a Sidon set. Then I is an independent set in $H^U(A)$.*

Proof. Otherwise, there exist $a_1, a_2 \in I$ and $u_1, u_2 \in U$ such that $a_1 + u_1 = a_2 + u_2$. Since we assume $I \cap U = \emptyset$, the elements $a_1, a_2 \in I$ and $u_1, u_2 \in U$ are distinct, which contradicts the fact that $U \cup I$ is a Sidon set. ■

1.1.4 Counting Sidon sets

The idea is the following: first, we choose a Sidon set U of size $\frac{\sqrt{n}}{\log n}$. Now, if $A = U \cup I$ is a Sidon set, then $I \subset V(H^U([n] - U))$ is an independent set by Proposition 1.1.7. This implies that

$$S(n) \leq \# \text{choices for } U \cdot \# \text{independent sets in } H^U([n] - U).$$

The reason to choose the size of U is that in the above inequality, we need to make sure that the number of choices for U is at most $2^{O(\sqrt{n})}$.

Application of Basic Graph Containers. First, let us try to apply the standard one-phase graph container method. Fix an arbitrary set $I \subseteq [n]$ such that $U \cup I$ is a Sidon set. We start with $A = [n] - U$ and $T = \emptyset$. In each iteration, we take a vertex v of maximum degree in $H^U(A)$. If $v \in I$, then we let $T \leftarrow T \cup \{v\}$, and $A \leftarrow A - \{v\} - N(v)$. Otherwise, let $T \leftarrow T$, and $A \leftarrow A - \{v\}$. We stop the algorithm when A is small enough, i.e. $A = O(\sqrt{n})$.

There are some issues with this argument. First, note that this method works only if $H^U(A)$ has many edges. By supersaturation, i.e. Lemma 1.1.5, we would require $|A||U| \geq 6n$, that is, $|A| \geq 6\sqrt{n} \log n$. Therefore, the size of A cannot reach $O(\sqrt{n})$. Moreover, we need the size of T to be at most $O\left(\frac{\sqrt{n}}{\log n}\right)$. It turns out that we can prove that $|T| \leq \frac{\sqrt{n}}{\log n}$ only when $|A| \geq 6\sqrt{n} \log^3 n$. This is because when A is small, the maximum degree of $H^U(A)$ also becomes small, and then we need to add more vertices in T to shrink the set A . The standard one-phase graph container method fails to work. Instead, we use a multi-phase algorithm to prove $S(n) = 2^{O(\sqrt{n})}$.

Proof. Fix a Sidon set I . Let U be an arbitrary subset of I with $|U| = \frac{\sqrt{n}}{\log^2 n}$.

Phase I(a). We start with $A = [n] - U$ and $T = \emptyset$. We shall apply the graph container algorithm on the graph $H^U(A)$ in the range $|A| \geq 6\sqrt{n} \log^6 n$.

To be more precise, in each iteration, we take a vertex $v \in A$ of maximum degree in $H^U(A)$. If $v \in I$, then we let $T \leftarrow T \cup \{v\}$, and $A \leftarrow A - \{v\} - N(v)$. Otherwise, let $T \leftarrow T$, and $A \leftarrow A - \{v\}$. We stop the algorithm when the size of A reaches $6\sqrt{n} \log^6 n$.

At the end of Phase I(a), let $T_I := T$, and $A_I := A$. By Lemma 1.1.5 and $|A| \geq 6\sqrt{n} \log^6 n$, in each iteration, we obtain that the average degree of $H^U(A)$ is

$$d(H^U(A)) = \frac{2e(H^U(A))}{|A|} \geq \frac{|A||U|^2}{6n} \geq |U| \log^4 n.$$

Therefore, we have $|T_I| \leq \frac{n}{|U| \log^4 n} = \frac{\sqrt{n}}{\log^2 n}$.

Phase I(b). We continue with $A = A_I$ and $T = \emptyset$. We run the graph container algorithm on $H^U(A)$ while $|A| \geq 6\sqrt{n} \log^2 n$. At the end of Phase I(b), let $T'_I := T$ and $A'_I := A$. Again, by Lemma 1.1.5 and $|A| \geq 6\sqrt{n} \log^2 n$, during the process, we have

$$d(H^U(A)) = \frac{2e(H^U(A))}{|A|} \geq \frac{|A||U|^2}{6n} \geq |U|.$$

Thus $|T'_I| \leq \frac{|A_I|}{|U|} \leq \frac{6\sqrt{n} \log^6 n}{|U|} = 6 \log^8 n$.

Phase II. We continue with $A = A'_I$ and $T = \emptyset$, and choose new $U_{II} \subset A \cap I$ with $|U_{II}| = \frac{\sqrt{n}}{\log n}$. We shall apply the graph container algorithm on the graph $H^{U_{II}}(A)$ until $|A| < 6\sqrt{n} \log n$.

At the end of Phase II, let $T_{II} := T$, and $A_{II} := A$. Note that in this phase, we still keep $|A||U_{II}| \geq 6n$, and thus $d(H^{U_{II}}(A)) \geq \frac{|A||U_{II}|^2}{6n} \geq |U_{II}|$. Therefore, $|T_{II}| \leq \frac{|A'_{II}|}{\frac{\sqrt{n}}{\log n}} = 6 \log^3 n$.

Phase III_j, for $1 \leq j \leq \log \log n$. For $j = 1$, we start with $A = A_{II}$ and $T = \emptyset$; for $j \geq 2$, we start with $A = A_{j-1}$ and $T = \emptyset$.

Let U_j be an arbitrary subset of $A \cap I$ of size $2^j \frac{\sqrt{n}}{\log n}$ and $A \leftarrow A - U_j$. We shall apply the graph container algorithm on the graph $H^{U_j}(A)$ in the range $\frac{6\sqrt{n} \log n}{2^{j-1}} > |A| \geq \frac{6\sqrt{n} \log n}{2^j}$.

At the end of Phase III_j, let $T_j := T$, and $A_j := A$. Note that in this phase, we still keep $|A||U_j| \geq 6n$, and similarly, we have $|T_j| \leq \frac{6\sqrt{n} \log n}{2^{j-1}} / 2^j \frac{\sqrt{n}}{\log n} = 3 \log^2 n$.

From the multi-phase graph container algorithm, for each Sidon set I , we build a certificate $\{U, U_{II}, U_1, \dots, U_{\log \log n}\} \cup \{T_I, T'_I, T_{II}, T_1, \dots, T_{\log \log n}\}$, which uniquely determines a set sequence $\{A_I, A'_I, A_{II}, A_1, \dots, A_{\log \log n}\}$. Let $C = A_{\log \log n}$. Then the following conditions are satisfied:

- $|U| = \frac{\sqrt{n}}{\log^2 n}$, $U_{II} = \frac{\sqrt{n}}{\log n}$;
- $|U_j| = 2^j \frac{\sqrt{n}}{\log n}$, $|A_j| \leq \frac{4\sqrt{n} \log n}{2^{j-1}}$, and $U_j \subseteq A_{j-1}$;
- $|T_I| \leq \frac{\sqrt{n}}{\log^2 n}$, $|T'_I| \leq 6 \log^8 n$, $|T_{II}| \leq 6 \log^3 n$, and $|T_j| \leq 3 \log^2 n$ for all $j \geq 1$;
- $I \subseteq C \cup T_I \cup T'_I \cup T_{II} \cup \bigcup_{j=1}^{\log \log n} T_j$, and $|C| \leq 6\sqrt{n}$.

Note that the third condition indicates that the number of choices for $\{T_I, T'_I, T_{II}, T_1, \dots, T_{\log \log n}\}$ is $2^{o(\sqrt{n})}$. The number of choices for U_I is $\binom{n}{\sqrt{n}/\log^2 n} = 2^{O(\sqrt{n}/\log n)}$, while the number of choices for $U_{II} \subset A'_I$ is $\binom{6\sqrt{n} \log^2 n}{\sqrt{n}/\log n} = 2^{O(\log \log n) \frac{\sqrt{n}}{\log n}}$. The next Claim shows that the number of choices for $\{U_1, \dots, U_{\log \log n}\}$ is $2^{O(\sqrt{n})}$.

Claim 1.1.8.

$$\prod_{j=1}^{\log \log n} \binom{\frac{6\sqrt{n} \log n}{2^{j-1}}}{2^j \frac{\sqrt{n}}{\log n}} \leq 2^{(2 \log(12e)+4)\sqrt{n}}.$$

Proof. Let $x = \frac{6\sqrt{n} \log n}{2^{\log \log n - 1}} = 12\sqrt{n}$. Then the left side is equal to

$$\begin{aligned} \prod_{i=0}^{\log \log n - 1} \binom{2^i x}{\frac{12n}{2^i x}} &\leq \prod_{i=0}^{\log \log n - 1} \left(\frac{e \cdot x^{2^{2i}}}{12n} \right)^{\frac{12n}{2^i x}} = \prod_{i=0}^{\log \log n - 1} (12e 2^{2i})^{\frac{\sqrt{n}}{2^i}} \\ &= \prod_{i=0}^{\log \log n - 1} 2^{[(\log(12e)+2i)2^{-i}]\sqrt{n}} \leq 2^{[\sum_{i=0}^{\infty} (\log(12e)+2i)2^{-i}]\sqrt{n}} \\ &= 2^{(2 \log(12e)+4)\sqrt{n}}, \end{aligned}$$

where the first inequality follows from $\binom{n}{k} \leq \left(\frac{en}{k}\right)^k$. ■

Therefore, the number of Sidon sets is at most

$$\begin{aligned} 2^{|C|} \cdot 2^{O(\sqrt{n}/\log n)} \cdot 2^{o(\sqrt{n})} \cdot 2^{O(\log \log n) \frac{\sqrt{n}}{\log n}} \cdot \prod_{j=1}^{\log \log n} \binom{|A_{j-1}|}{2^j \frac{\sqrt{n}}{\log n}} \\ = 2^{O(\sqrt{n})} \cdot \prod_{j=1}^{\log \log n} \binom{\frac{6\sqrt{n} \log n}{2^{j-1}}}{2^j \frac{\sqrt{n}}{\log n}} = 2^{O(\sqrt{n})}, \end{aligned}$$

where the number of containers is bounded by the number of choices for $T_I, T'_I, T_{II}, T_1, \dots, T_{\log \log n}, U, U_{II}, U_1, \dots, U_{\log \log n}$, respectively. ■

1.2 Improving the constant

In this section, we present the proof in [8] to get a better constant. The main different idea is that: we will stop the iteration when applying the container lemma if the certificate is big enough. More precisely, we will use the following lemma.

Lemma 1.2.1. *Let G be a graph on N vertices, let q be an integer and let $0 \leq \beta \leq 1$ and R be real numbers with*

$$R \geq e^{-\beta q} N.$$

Suppose that the number of edges $e(U)$ induced in G by any subset $U \subset V(G)$ with $|U| \geq R$ satisfies

$$e(U) \geq \beta \binom{|U|}{2}.$$

Then, for all integers $r \geq 0$, the number of independent sets in G of cardinality $q + r$ is at most $\binom{N}{q} \binom{R}{r}$.

Proof. Fix an independent I in G of size $q + r$. We start the container algorithm with $A = V(G)$ and $T = \emptyset$. We will stop when the certificate T has size q . In the end, we obtain (T, A) such that

$T \subset I \subset (T \cup A)$. Using the assumption, we can prove that $|A| \leq (1 - \beta)^q N \leq R$. Hence, we obtain the conclusion by counting the number of choices for T and A at the end of the algorithm. Some details are needed, but we omit them. ■

Let $Z_n(t)$ be the family of Sidon sets of cardinality t contained in $[n]$. The following lemma, viewed as a corollary of Lemma 1.2.1, gives us a recursive bound for the number of Sidon sets.

Lemma 1.2.2. *Let n, s and q be integers and let $0 < \sigma < 1$ be a real number such that*

$$\frac{s^2 q}{n} \geq \frac{2}{1 - \sigma} \log \frac{s\sigma}{2}.$$

Then, for any integer $r \geq 0$, we have

$$|Z_n(s + q + r)| \leq |Z_n(s)| \binom{n}{q} \binom{2n/\sigma s}{r}.$$

Proof. Let $S_0 \subset [n]$ be an arbitrary Sidon set with $|S_0| = s$. We show that the number of Sidon sets $S \subset [n]$ with $S_0 \subset S$ and $|S| = s + q + r$ is at most $\binom{n}{q} \binom{2n/\sigma s}{r}$ whence our lemma will follow.

To do this, we apply Lemma 1.2.1 for $H^{S_0}([n] - S_0)$ with $\beta = (1 - \sigma)s^2/2n$ and $R = 2n/\sigma s$. We can easily prove the edge-density condition in Lemma 1.2.1 for $U \subset ([n] - S_0)$ with $|U| \geq R$ by following the proof of Lemma 1.1.5. Hence, we finish the proof. ■

Now, we will derive the main theorem by applying Lemma 1.2.2.

Theorem 1.2.3 (Kohayakawa-Lee-Rödl-Samotij [8]). *Let $0 < \sigma < 1$ be a real number. For any large enough n and $t \geq 2s_0$, where $s_0 = \left(\frac{2}{1-\sigma} n \log n\right)^{1/3}$, we have*

$$|Z_n(t)| \leq n^{3s_0} \left(\frac{32en}{\sigma t^2}\right)^t.$$

Proof. Fix integers n and t with $2s_0 \leq t \leq (1 + o(1))\sqrt{n}$. Let K be the largest integer satisfying $t2^{-K} \geq 2s_0$. We define three sequences $(s_k), (r_k), (q_k)$ for $0 \leq k \leq K$ as following: $q_0 = s_0, r_0 = t2^{-K} - s_0 - q_0$; $s_1 = t2^{-K}, q_1 = q_0/4, r_1 = t2^{-K+1} - r_0 - q_1$; and for $k \geq 2$,

$$\begin{cases} s_k &= 2s_{k-1} = t \cdot 2^{-K+k-1}, \\ q_k &= q_{k-1}/4 = 4^{-k}q_0, \\ r_k &= \frac{t}{2^{K-k}} - s_k - q_k. \end{cases}$$

Since $s_k^2 q_k \geq s_0^2 q_0 = 2(1 - \sigma)^{-1} n \log n > 2(1 - \sigma)^{-1} n \log(\sigma s_k/2)$ for all $0 \leq k \leq K$, we may apply Lemma 1.2.2 recursively to obtain

$$|Z_n(t)| \leq |Z_n(s_0)| \prod_{0 \leq k \leq K} \binom{n}{q_k} \prod_{0 \leq k \leq K} \binom{2n/\sigma s_k}{r_k}.$$

Note that $|Z_n(s_0)| \leq \binom{n}{s_0} \leq n^{2s_0/3}$ and that

$$\prod_{0 \leq k \leq K} \binom{n}{q_k} \leq n^{\sum_{0 \leq k \leq K} q_k} \leq n^{q_0 \sum_{0 \leq k \leq K} 1/4^k} \leq n^{4q_0/3} = n^{4s_0/3}.$$

The remaining is to estimate $\prod_{0 \leq k \leq K} \binom{2n/\sigma s_k}{r_k}$. By the choice of K , we have $(r_0 + s_0 + q_0)/2 = t \cdot 2^{-K-1} < 2s_0$. Therefore, we have $\binom{2n/\sigma s_0}{r_0} \leq n^{s_0}$ for all large n . We now note that

$$\begin{aligned} \prod_{1 \leq k \leq K} \binom{2n/\sigma s_k}{r_k} &= \prod_{1 \leq k \leq K} \binom{2n/\sigma s_{K-k+1}}{r_{K-k+1}} \leq \prod_{1 \leq k \leq K} \binom{2n/\sigma s_{K-k+1}}{r_{K-k+1} + q_{K-k+1}} \\ &= \prod_{1 \leq k \leq K} \binom{2n/\sigma t 2^{-k}}{t 2^{-k}} \leq \prod_{1 \leq k \leq K} \left(\frac{2^{2k+1} en}{\sigma t^2} \right)^{t 2^{-k}} \\ &\leq \left(\frac{2en}{\sigma t^2} \right)^{t \sum_{k \geq 1} 2^{-k}} 2^{2t \sum_{k \geq 1} k 2^{-k}} = \left(\frac{2en}{\sigma t^2} \right)^t 2^{4t} = \left(\frac{32en}{\sigma t^2} \right)^t. \end{aligned}$$

Thus, we complete the proof of Theorem 1.2.3. ■

Proof of Theorem 1.1.4. Let $\sigma = 32/33$ in Theorem 1.2.3, then $s_0 = (66n \log n)^{1/2}$. For large n , we have

$$\begin{aligned} S(n) &= \sum_{0 \leq t \leq (1+o(1))\sqrt{n}} |Z_n(t)| = \sum_{t < 2s_0} |Z_n(t)| + \sum_{2s_0 \leq t \leq (1+o(1))\sqrt{n}} |Z_n(t)| \\ &\leq \sum_{t < 2s_0} \binom{n}{t} + \sum_{2s_0 \leq t \leq (1+o(1))\sqrt{n}} \left(\frac{33en}{t^2} \right)^t \leq n^{2s_0} + \sqrt{n} \cdot (33e)^{(1+o(1))\sqrt{n}} \leq 2^{c\sqrt{n}} \end{aligned}$$

for any constant $c > \log(33e)$, where the last inequality is followed by the fact that $f(t) = (33en/t^2)^t$ is increasing on the interval $(0, \sqrt{33n/e})$. ■

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